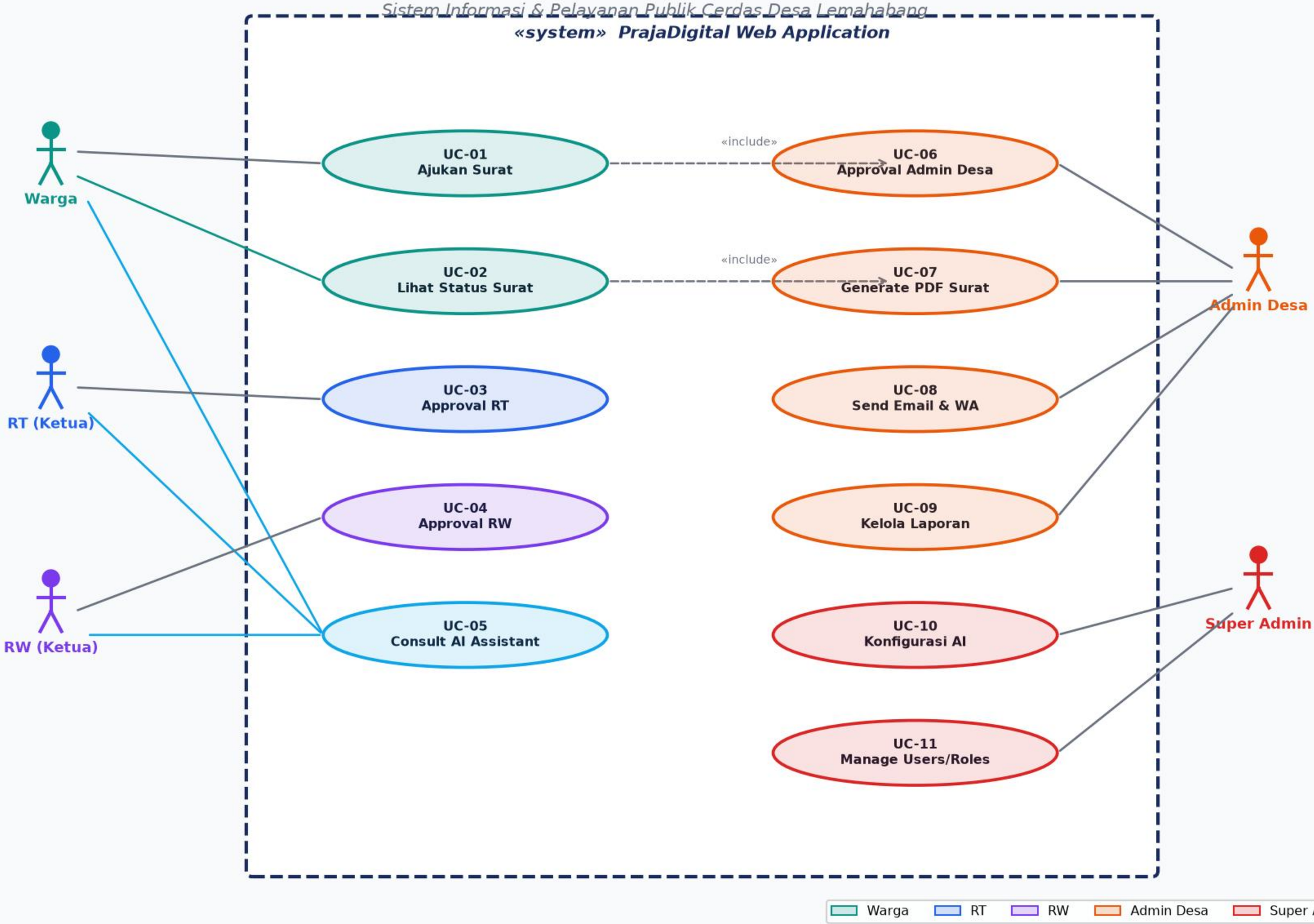
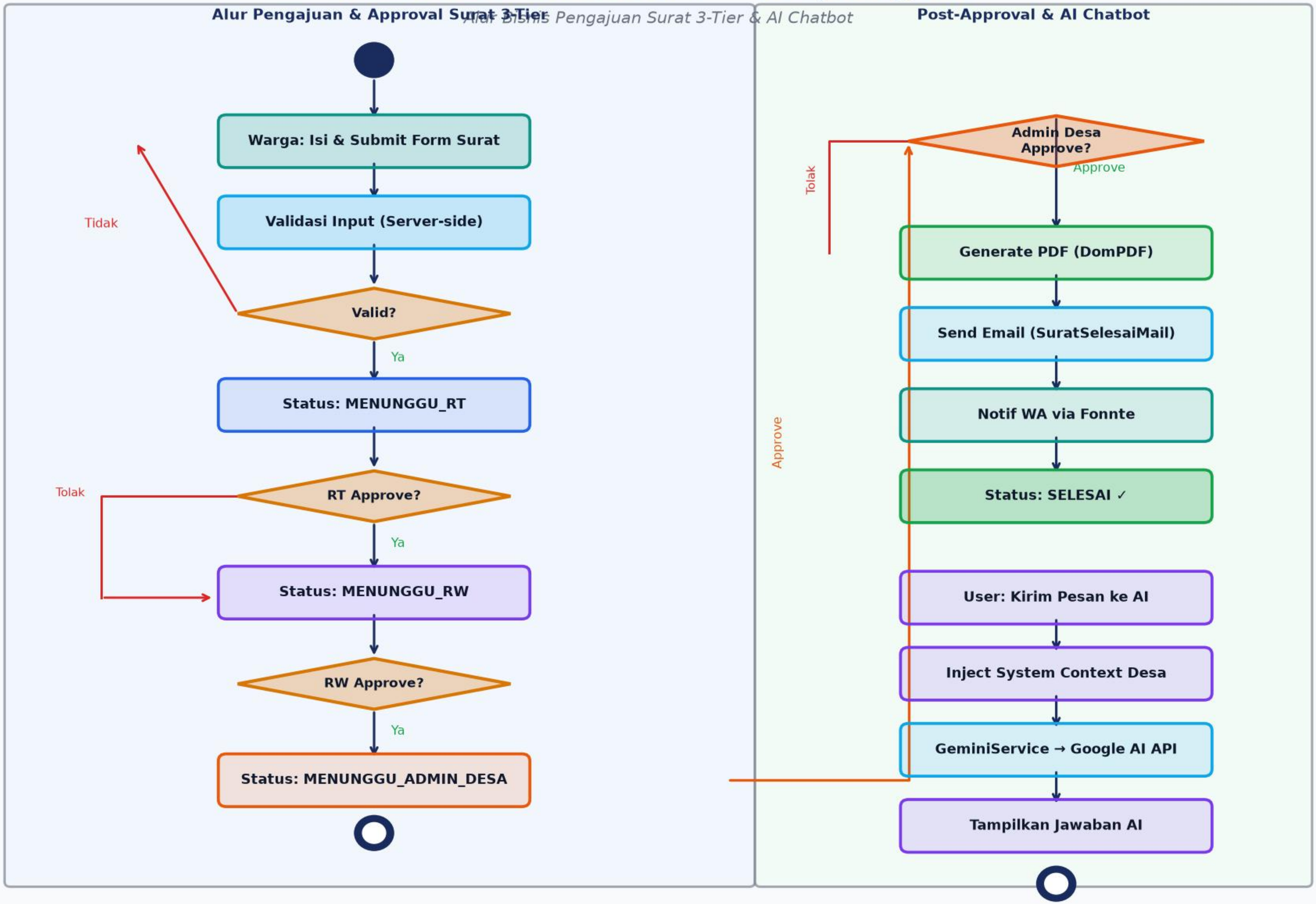
USE CASE DIAGRAM — PrajaDigital

Sistem Informasi & Pelayanan Publik Cerdas Desa Lemahabang



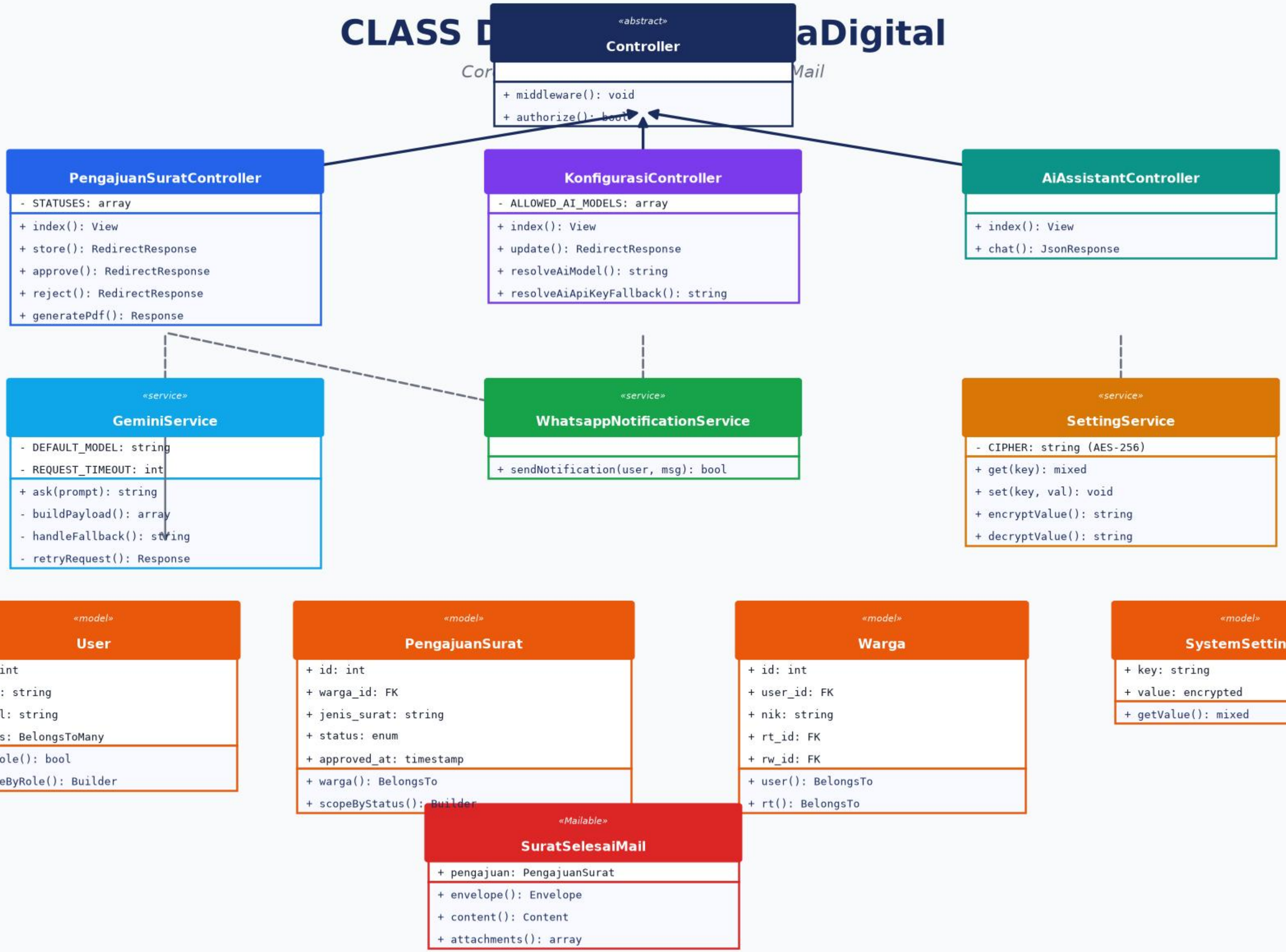
ACTIVITY DIAGRAM — PrajaDigital

Alur Bisnis Pengajuan Surat 3-Tier & AI Chatbot



CLASS D aDigital

Cor Mail



SEQUENCE DIAGRAM — PrajaDigital

Surat → PDF → App

Admin Desa (Browser) | Middleware (Auth+Role) | PengajuanSurat Controller | PengajuanSurat Model | MySQL DB | DomPDF Service | Mail + Fonnte WA

1. POST /pengajuan/{id}/approve
2. forward request (Auth OK)
3. find(id)
4. SELECT * WHERE id=:id
5. *PengajuanSurat instance*
6. *return $pengajuan*
7. update(status=SELESAI, approved_at)
8. UPDATE pengajuan_surat SET ...
9. *1 row affected*
10. generatePdf($pengajuan)
11. *PDF binary (base64)*
12. Mail::to()->send(SuratSelesaiMail)
13. WhatsappService::send(notif)
14. *true (sent OK)*
15. *redirect()->back()->with('success',...)*

ENTITY RELATIONSHIP DIAGRAM — PrajaDigital

Foreign Key

users
- [PK] id INT PK
- name VARCHAR(255)
- email VARCHAR(255)
- password VARCHAR(255)
- created_at TIMESTAMP

roles
- [PK] id INT PK
- name VARCHAR(125)
- guard_name VARCHAR(125)

model_has_roles
- [FK] role_id INT FK
- model_type VARCHAR
- [FK] model_id INT FK

warga
- [PK] id INT PK
- [FK] user_id INT FK
- nik VARCHAR(16)
- nama VARCHAR(255)
- [FK] rt_id INT FK
- [FK] rw_id INT FK
- [FK] desa_id INT FK

rt
- [PK] id INT PK
- nama_rt VARCHAR(10)
- [FK] ketua_id INT FK
- [FK] rw_id INT FK

rw
- [PK] id INT PK
- nama_rw VARCHAR(10)
- [FK] ketua_id INT FK
- [FK] desa_id INT FK

desa
- [PK] id INT PK
- nama_desa VARCHAR(255)
- kecamatan VARCHAR(100)
- kabupaten VARCHAR(100)

pengajuan_surat
- [PK] id INT PK
- [FK] warga_id INT FK
- jenis_surat VARCHAR(100)
- status ENUM
- data_tambahan JSON
- approved_rt_at TIMESTAMP
- approved_rw_at TIMESTAMP
- approved_admin_at TIMESTAMP
- pdf_path VARCHAR(255)
- created_at TIMESTAMP

surat_templates
- [PK] id INT PK
- jenis_surat VARCHAR(100)
- template_path VARCHAR
- is_active BOOLEAN

system_settings
- [PK] id INT PK
- key VARCHAR(100)
- value TEXT encrypted
- updated_at TIMESTAMP

audit_logs
- [PK] id INT PK
- [FK] user_id INT FK
- action VARCHAR(255)
- model_type VARCHAR
- model_id INT
- created_at TIMESTAMP

permissions
- [PK] id INT PK
- name VARCHAR(125)
- guard_name VARCHAR(125)

role_has_permissions
- [FK] permission_id INT FK
- [FK] role_id INT FK